

विश्वकर्मा
— PRESENTS —
FARLANDS

**PROBLEM
STATEMENTS**

9 REAL-WORLD
INNOVATION MISSIONS



Every mission is designed to allow teams to start with a strong foundation, build a functional prototype, and then explore how far they can take their idea within the hackathon.

How the Missions Work

Stage 1: Foundation Minimum Viable Solution

Stage 2: Functional Solution Complete Working Prototype

Stage 3: Innovation and Intelligence Advanced Exploration

Stage 3 is not compulsory. A strong and well-executed Stage 1 or Stage 2 solution can still be competitive

Innovation Mission 1

IMU Integration: The Smartphone as a High-Fidelity Controller

Challenge Brief

Problem: Explore how smartphone motion sensors can be used as an intuitive control interface.

Build: Create a system where smartphone motion and orientation control a physical, simulated, or virtual system.

Stage 1: Foundation

Establish communication using smartphone motion sensors. Interpret accelerometer or gyroscope data and demonstrate basic motion-based control.

Stage 2: Functional Solution

Develop a working application where real-time smartphone sensor data controls a robotic, simulated, or virtual system with stable and responsive interaction.

Stage 3: Innovation and Intelligence

Enhance the system through additional sensor integration, improved control logic, intelligent behaviour, precision control, or a novel application.

Minimum Deliverable

Demonstrate real-time smartphone motion controlling a physical, simulated, or virtual system.

Constraints

Teams may use physical hardware or simulation platforms.

Suggested Technologies

Smartphone sensors, ESP32, Arduino, Bluetooth, Wi-Fi, Unity, Unreal Engine, or equivalent platforms.

Judging Focus

Accuracy of motion control, responsiveness, functionality, creativity, and innovation.

Innovation Mission 2

Computer Vision and Virtual Logistics

Challenge Brief

Problem: Explore how visual information from the physical world can influence operations within a digital environment.

Build: Create a system that detects physical objects or visual markers and uses the information to control a virtual logistics workflow.

Stage 1: Foundation

Develop a system capable of detecting and identifying physical objects or visual markers using a camera.

Stage 2: Functional Solution

Use detected objects or markers to control a functional workflow involving movement, sorting, retrieval, or positioning of virtual assets.

Stage 3: Innovation and Intelligence

Introduce collision avoidance, route optimisation, automated decision-making, multi-object interaction, or dynamic environmental changes.

Minimum Deliverable

Demonstrate successful visual detection and its use in controlling or influencing a virtual logistics workflow.

Constraints

Physical hardware beyond a camera is not mandatory.

Suggested Technologies

Python, OpenCV, computer vision models, webcams, Unity, Unreal Engine, or equivalent platforms.

Judging Focus

Detection reliability, physical-to-virtual interaction, functionality, and innovation.

Innovation Mission 3

Voice-Controlled Smart Environment

Challenge Brief

Problem: Explore how natural voice interaction can control and manage connected environments.

Build: Create a system where voice commands interact with and control multiple physical or simulated components.

Stage 1: Foundation

Develop a system capable of performing Voice Activity Detection (VAD), receiving voice commands, and controlling at least one connected or simulated environment component.

Stage 2: Functional Solution

Expand the system to control multiple interconnected components based on user intent and the current state of the environment.

Stage 3: Innovation and Intelligence

Introduce contextual decision-making where previous interactions, environmental conditions, or multiple inputs influence the response.

Minimum Deliverable

Demonstrate Voice Activity Detection and voice commands controlling at least two connected or simulated components with visible system feedback.

Constraints

Physical devices are not compulsory. Simulation-based environments are permitted

Suggested Technologies

Voice Activity Detection (VAD), speech-to-text APIs, Python, ESP32, Arduino, Unity, Unreal Engine, or equivalent platforms.

Judging Focus

Voice recognition, VAD implementation, responsiveness, functionality, user experience, and intelligent automation.

Innovation Mission 4

Procedural Environments: Real-Time API Integration

Challenge Brief

Problem: Explore how real-world information can dynamically influence a virtual environment.

Build: Create an interactive environment that responds to real-time external data

Stage 1: Foundation

Connect a live environmental or weather data source to a virtual environment and demonstrate changes based on incoming information.

Stage 2: Functional Solution

Develop multiple environmental elements that respond dynamically to real-world data, such as weather, lighting, visibility, terrain conditions, or time-of-day cycles.

Stage 3: Innovation and Intelligence

Introduce interactions between multiple environmental systems or predictive behavior that allows intelligent adaptation to changing data patterns.

Minimum Deliverable

Demonstrate a virtual environment that visibly changes based on live or externally provided data.

Constraints

The use of physical hardware is not required.

Suggested Technologies

Weather APIs, external APIs, Python, Unity, Unreal Engine, procedural generation tools, or equivalent platforms.

Judging Focus

Data integration, dynamic behaviour, functionality, visual implementation, and innovation.

Innovation Mission 5

Cognitive NPCs: Conversational Dialogue Systems

Challenge Brief

Problem: Explore how game characters can move beyond predefined dialogue and respond naturally to player interaction.

Build: Create an interactive NPC capable of communicating with players and influencing the surrounding game experience.

Stage 1: Foundation

Develop an NPC capable of receiving and responding to natural language input from a player.

Stage 2: Functional Solution

Integrate the NPC into a functional game workflow where conversations influence quests, decisions, objectives, or player progression.

Stage 3: Innovation and Intelligence

Introduce memory, context awareness, dynamic dialogue, personality, or adaptive behavior based on player interactions.

Minimum Deliverable

Demonstrate an interactive NPC responding to player input and influencing at least one aspect of the application or game.

Constraints

Teams may use existing AI models, APIs, or open-source frameworks.

Suggested Technologies

LLMs, AI APIs, Python, Unity, Unreal Engine, conversational systems, memory systems, or equivalent platforms.

Judging Focus

Interaction quality, contextual behaviour, application integration, creativity, and innovation.

Innovation Mission 6

Disaster Response Drones: Predictive Search and Situational Awareness

Challenge Brief

Problem: During a disaster, identifying affected areas and deciding where response efforts should be prioritised can be difficult.

Build: Create a system that uses aerial imagery, environmental information, GIS data, geographical information, or other relevant inputs to support disaster response.

Stage 1: Foundation

Develop a system capable of processing aerial imagery, sensor data, and GIS data or geographical information to identify and visualise affected areas or potential hazards.

Stage 2: Functional Solution

Integrate multiple sources of information, including GIS data where relevant, to analyse disaster conditions and support decisions such as identifying priority zones, planning search routes, or allocating response resources.

Stage 3: Innovation and Intelligence

Introduce predictive analysis using GIS data, environmental, geographical, weather, or historical data to estimate high-risk regions or areas requiring priority attention.

Minimum Deliverable

Demonstrate a system capable of analysing disaster-related information, including GIS or geographical data, and identifying or prioritising areas requiring response.

Constraints

Teams may use simulated drone data, public datasets, recorded aerial imagery, GIS data, geographical information, or physical drones. A physical drone is not mandatory.

Suggested Technologies

Python, computer vision, GIS data and mapping tools, weather data, simulation platforms, machine learning models, or equivalent technologies.

Judging Focus

Situational analysis, effective use of GIS and multiple data sources, usefulness of predictions, functionality, and innovation.

Innovation Mission 7

Connected Infrastructure: Adaptive Intelligent IoT Systems

Challenge Brief

Problem: Traditional automation systems often react to individual sensor readings without understanding the overall condition of the environment.

Build: Create an interconnected IoT system where multiple sensing components work together to influence coordinated responses.

Stage 1: Foundation

Develop an IoT system consisting of multiple sensors and connected devices capable of collecting and communicating real-time data.

Stage 2: Functional Solution

Design the system so information from multiple sensors influences multiple connected components through coordinated responses rather than simple threshold-based automation.

Stage 3: Innovation and Intelligence

Introduce pattern analysis, prediction, anomaly detection, or adaptive decision-making for changing or unexpected conditions.

Minimum Deliverable

Demonstrate multiple sensors influencing coordinated responses across multiple connected components.

Constraints

Teams may use physical IoT hardware or simulated sensor data where appropriate.

Suggested Technologies

ESP32, Arduino, Raspberry Pi, DHT sensors, ultrasonic sensors, LDRs, MQTT, cloud platforms, Python, or equivalent technologies.

Judging Focus

Sensor integration, coordination, adaptive behaviour, functionality, and intelligent decision-making.

Innovation Mission 8

IoT Meets Robotics: Adaptive Remote Robotics

Challenge Brief

Problem: Remote robotic systems depend heavily on communication, but real-world environments may introduce delays, interruptions, and changing conditions.

Build: Create a connected robotic system capable of receiving commands, sharing real-time information, and responding intelligently to operational conditions.

Stage 1: Foundation

Implement a functional system where the robot receives remote commands while also responding to local sensor information and operational conditions.

Stage 2: Functional Solution

Design the system so information from multiple sensors influences multiple connected components through coordinated responses rather than simple threshold-based automation.

Stage 3: Innovation and Intelligence

Introduce adaptive behaviour for communication delays, connection loss, changing environmental conditions, or conflicting sensor and remote control inputs.

Minimum Deliverable

Demonstrate a robotic system capable of transmitting information and receiving remote control commands.

Constraints

Simulation is acceptable where physical robotic hardware is unavailable.

Suggested Technologies

ESP32, Arduino, Raspberry Pi, motors, motor drivers, ultrasonic sensors, Wi-Fi, Bluetooth, MQTT, robotics simulation platforms, or equivalent technologies.

Judging Focus

Reliability, remote and local control integration, resilience, functionality, and innovation.

Innovation Mission 9

Intelligent Robotics: Autonomous Object Interaction

Challenge Brief

Problem: Robotic systems must understand their environment before making decisions and interacting with objects.

Build: Create a robotic or simulated system capable of perceiving objects, making decisions, and performing appropriate actions.

Stage 1: Foundation

Develop a system capable of detecting or identifying objects using sensors, computer vision, or other perception methods.

Stage 2: Functional Solution

Enable the robot to make a decision and perform an appropriate physical or simulated action, such as sorting, retrieving, transporting, or repositioning an object.

Stage 3: Innovation and Intelligence

Introduce dynamic decision-making where the robot adapts to changing object positions, environmental conditions, multiple object types, or unexpected situations.

Minimum Deliverable

Demonstrate successful object perception followed by an appropriate physical or simulated interaction.

Constraints

Simulation is acceptable where physical robotic hardware is unavailable.

Suggested Technologies

ESP32, Arduino, Raspberry Pi, cameras, ultrasonic sensors, OpenCV, motors, servo motors, robotic platforms, simulation environments, or equivalent technologies.

Judging Focus

Object perception, decision-making, autonomous interaction, functionality, adaptability, and innovation.



ALL THE BEST !